

Deep Learning Approach for Medicinal Plant Identification and Therapeutic Use Prediction

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Abstract—The therapeutic nature of medicinal plants and their ability to heal many diseases raises the need for their automatic identification. However, the preservation and transmission of traditional medicinal knowledge encounter substantial problems due to factors such as professional secrecy, wrong cultural beliefs and lack of experts on the field. Consequently, there is a pressing risk of traditional medicinal knowledge fading away, resulting in a decline in expertise and an increased likelihood of errors in a field that demands serious attention and care. This work explores deep learning and proposes a deep learning approach for medicinal plant identification and to predict therapeutic use of medicinal plants from their leaf image. The transfer learning approach along with the pre-trained neural networks such as VGG16, InceptionV3 and Xception architectures were employed to extract features from the input leaf images. The Softmax and support vector machine(SVM) were used to classify the features generated by the three feature extractor model. The InceptionV3 architecture outperformed the state-of-the-art pre-trained models by achieving 99.17% average validation accuracy of Support Vector Machine (SVM) and softMax classifier) and the maximum validation accuracy of

99.34 is obtained using SVM. The average validation accuracy of the two classifier on each feature extractor model is used to select the better feature extractor model. To improve performance the SVM classifier was tuned using the Bayesian optimization technique and the proposed model InceptionV3 was validated with our custom datasets and achieved 99.34% prediction accuracy. The finding of this work showcases transfer learning-based InceptionV3 feature with support vector machine classifier(SVM), can effectively predict the therapeutic use of medicinal plants from leaf images with high accuracy and offers a promising solution to the fading traditional medicinal knowledge.

Index Terms—CNN, deep learning, SVM, TM .

I. INTRODUCTION

According to the World Health Organization (WHO), traditional medicine is a set of health practices, knowledge, approaches, and beliefs obtained by combining medicinal plants, mineral-based medicines, animals, spiritual therapies, and manual techniques to diagnose, prevent, and treat diseases

[1]. In most cases, traditional medicine(TM) uses medicinal plants that are used to treat a variety of common and chronic diseases in different Asian and African countries [2]. Ethiopia is a home to more than 80 ethnic groups with a diversified culture and traditional knowledge, which are reflected in the use of herbal medicine. According to [3] traditional medicinal knowledge is dynamic and diverse across different locations, regions, and linguistic and cultural backgrounds, each having their own unique knowledge about medicinal plant usage, preparation, and conservation. This knowledge persists as secret oral traditions in the indigenous community. This leads to traditional medicinal knowledge is only known by a few experts, mostly rural and indigenous people, and there is no easy access to its evidence. Therefore, promoting traditional medicine knowledge to the general public and researchers is essential to preserve traditional medicine knowledge and to maintain a healthy herbal lifestyle [4]. In this work, we propose a deep learning approach for the medicinal plant identification and therapeutic use prediction of indigenous medicinal plants in Ethiopia to maintain medicinal knowledge.

II. LITERATURE REVIEW

Plant identification has always been the subject of a large variety of studies in many fields, such as medicine, industry, architecture, agronomy, and so on [5]. In recent years, the identification and correct classification of medicinal plants have become a hot research area and have gained the interest of various researchers.

Dileep M.R. and Pournami P.N. [6] proposed a deep learning-based customized convolutional neural network whose architecture was designed based on Alexnet's architecture for the classification of medicinal plants. In this work, leaf features such as shape, size, color, and texture were applied, and 2400 images from 40 medicinal plant species were collected as a dataset. After the proposed model's implementation, 96.76% accuracy was achieved. But in this work, the size and diversity of the datasets are very small, which leads to non-ideal robustness of the model, and it needs some techniques.

Amuthalingeswaran et al. [7] proposed a custom-designed CNN model called the Medicinal Neural Networks (MNN) for the identification of medicinal plants. Five convolutional layers, four max pooling layers and 3x3 filters with a stride value of 1 were utilized in MNN model. 8259 images that belong to four classes of medicinal plant species were used to train and test the MNN model. Finally, 85 percent testing accuracy was achieved. But this accuracy needs to be further improved.

Anu Paulson and Ravishankar [8] proposed AI based indigenous medicinal plant identification system using a custom design CNN and pre-trained VGG16 and VGG19 deep learning methods. 64 classes of medicinal plant leaves with 1000 samples per each class were utilized for training and testing purpose. Finally they proposed VGG16 as a best model with its classification accuracy 97.8%. However the paper does not mention the model complexity that have a crucial impact to deploy it in real-time application.

Mukherjee et al. [2] suggested a CNN-driven computer vision framework for the identification of medicinal leaf species and their maturity stages. The method used three convolutional layers followed by a max pooling layer after each convolution to extract features of medicinal leaves. A fully connected layer with 1024 nodes and the binary particle swarm optimization (BPSO) technique were used to build their model. To train the CNN model, the work used 4500 images of medicinal plant leaves that belong to three classes. Finally, the CNN model with the binary particle swarm optimization (BPSO) technique achieve 99% test accuracy.

S. Roopashree, (Member, IEEE), AND J. Anitha [9] proposed a vision-based system for medicinal plant identification using Xception features by combining deep learning and computer vision techniques to develop the DeepHerb model. The model was trained on a dataset comprising 2515 images of 40 class of medicinal herbs, collected from different regions of Karnataka in southern India. To extract discriminating features, the authors utilized pre-trained VGG16, VGG19, InceptionV3, and Xception CNN models with their pre-optimized weights by cutting their top layers. The extracted

features are then classified using Artificial Neural Network (ANN) and Support Vector Machine (SVM) techniques. As a result, the performance of the DeepHerb dataset was evaluated by comparing the accuracy of the different pre-trained models and the results demonstrate that the DeepHerb model, trained on Xception features and using ANN, outperforms the other models with a testing accuracy of 97.5%.

Silky Sachar and Anuj Kumar [10] proposed an ensemble learning approach that combines the outputs of three component models, such as MobileNetV2, InceptionV3, and ResNet50 to achieve accurate medicinal leaf identification. The training process involves extracting features from the input leaf images using the pre-trained models and using a softmax layer connected to a dense layer as the classifier. The accuracy of the component models were evaluated using three-fold and five-fold cross-validation. Finally, an ensemble model was constructed by taking the weighted average of the outputs of the component models and outperforms the individual pre-trained models by achieving 99.66% accuracy on the test set. However, ensemble techniques, are computationally expensive and require substantial computational resources.

III. MATERIALS AND METHODS

A. Sample Collection

In this work, 27 indigenous medicinal plant species were selected based on their medicinal use and popularity among the Ethiopian community. The samples were collected from four different geographical locations such as Gojjam, Jimma, Awi and Gondar to include the impact of the location on leaf feature. During sample collection both the top and bottom sides of the medicinal plant leaves were considered to extract more features in the feature extraction phase. The sampled leaf datasets consists of 200 samples per class, with 100 sample images at the top and 100 sample images at the bottom.

B. Image Labeling

Since the proposed system is based on a supervised learning approach, all categories of medicinal

leaf images should be labeled with their corresponding therapeutic uses and their species name separately. As a result, labeling tasks were performed with the help of i) literature conducted to document medicinal plants and their corresponding medicinal use [3] and ii) with the help of four traditional medicinal practitioners, working at Arsema traditional medication, Jerusalem traditional medication, Tsimret homemade traditional medication, and Etse-hiwote traditional medication. With the help of previous works and four medicinal experts who are currently working on traditional medicine preparation, 27 species of medicinal plant leaf images were labeled into their corresponding therapeutic use. To maintain confidentiality about the therapeutic uses of medicinal plants, medicinal plant leaf images were labeled with the help of previous studies and four traditional medicinal practitioners. After the labeling process, medicinal plant species that have been labeled similarly by at least three medicinal practitioners and the previous studies were taken.

C. Image Pre-processing

The most important problem in CNN-based DL's application of image classification is insufficient datasets in size and diversity. To improve the insufficient datasets problem of the custom datasets, an image augmentation technique was applied using an image generator object. Furthermore, image resizing and normalization techniques were also applied to reduce computational time.

D. Feature Extraction

In computer vision problems, there are a variety of features for neural networks to learn about in the training dataset. Among them, edges, corners, round shapes, curvy shapes, and blobs are low-level features, while eyes, circles, squares, and wheels are mid- and higher-level features. There are many details in the images that CNN can pick up on. But if we have only small images in our training dataset, this may not be enough data for the model to learn all those things. Hence, utilizing a network that has already been trained is a common and efficient method for deep learning approaches on

small datasets. So in this work, pre-trained deep learning networks were used as feature extractors, to overcome the impact of small data size on the model's performance.

1) *VGG16*: VGG16 is a convolutional neural network that has been trained on a dataset of over one million images from the ImageNet database. It features 16 weight layers in its configuration, capable of classifying images into 1000 different object categories. The network comprises 13 convolutional layers and 3 fully connected layers, totaling 16 weight layers. The final layers include two fully connected layers, each containing 4096 units, and a third layer with 1000 units, enabling 1000-way classification. The softmax layer serves as the concluding layer for classification purposes.

2) *InceptionV3*: InceptionV3 is one of the deep learning models that was designed based on convolutional neural networks to be used in image classification tasks. It uses less computational power and proved to be more efficient in terms of network parameters and the costs associated with memory requirements or other resources compared to the previous version. The novel idea in the architecture includes factorized convolutions to test network efficiency, smaller convolutions to make training faster, asymmetric convolutions, an auxiliary classifier to regularize the network, and pooling operations to reduce the grid size [10].

3) *Xception*: The vital concept of the Xception network is its depth-wise separable convolutions and residual connections. It is stimulated by the Inception architecture, where the modules in Inception are replaced by depth-wise separable convolutions [11]. Here, the original depthwise convolution in the Inception model was replaced by changing the order of operations (i.e., in the Inception model, channel-wise spatial convolution is performed first and then point-wise convolution), whereas in the Xception model, 1×1 (pointwise convolution) is performed and followed by channel-wise spatial convolution ($n \times n$) to reduce computational complexity. The Xception model consists of 36 convolution layers structured into 14 modules and three major flows known as entry flow, middle flow, and exit flow. These axes are subsequently used as point

wise convolution for performing cross-channel correspondence.

IV. RESULT AND DISCUSSION

A. Feature Extractor Model Performance

To determine why the Xception model was selected as the base model for the given dataset, the extracted features were evaluated using the softmax classifier and SVM with the default hyperparameter values. Finally, the evaluation result with precision, recall, accuracy and loss metrics is used as a base model selection criterion. As shown in Table I and II, the Xception model performed better than the other in the softmax classifier in terms of accuracy, precision, recall, and loss. In terms of accuracy, Xception scored 99.2% validation accuracy. The InceptionV3 model followed with 99.0%. The VGG16 feature extractor, on the other hand, achieved 95.3% when validated with validation set of our custom dataset, which is lower than the Xception and InceptionV3 feature extractors. the lower accuracy in VGG16 ensures that the VGG16 features are not appropriate for this particular problem compared to Xception, and Inception features. The accuracy and loss graph of the three feature extractor model using softmax as a classifier is shown in Figure 3.

TABLE I: Feature extractor performance using softmax classifier on the validation set

Model	Precision	Recall	Accuracy	Loss
VGG16	99	77	95.3	0.04
InceptionV3	99.2	98.4	99	0.07
Xception	99.2	98.7	99.2	0.06

TABLE II: Feature extractor performance using softmax classifier on the training set

Model	Precision	Recall	Accuracy	Loss
VGG16	98.8	70	92.0	0.01
InceptionV3	98.3	96.2	97.6	0.02
Xception	99.4	98.0	99.2	0.09

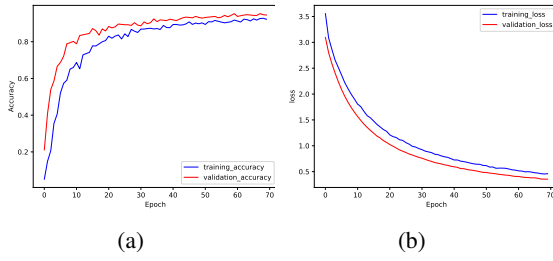


Fig. 1: (a) VGG16 accuracy (b) VGG16 loss

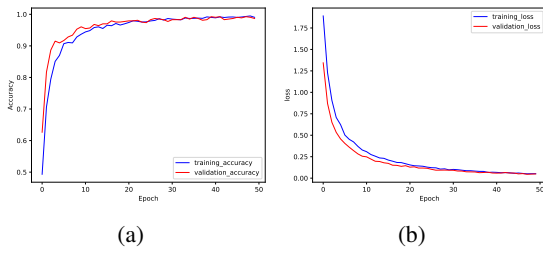


Fig. 2: (a) Xception accuracy (b) Xception loss

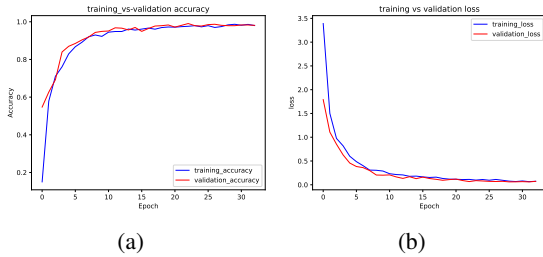


Fig. 3: (a) InceptionV3 accuracy (b) InceptionV3 loss

For the sake of model robustness and to check the features' dependability on the classifiers, we also classify the extracted deep learning features using the SVM classifier, and the resulting accuracy is shown in Table III. As shown in the table, InceptionV3 performs as a better feature extractor for appropriately predict the therapeutic use of medicinal leaves. Its higher SVM accuracy on both training (99.9%) and validation data (99.34%) compared to Xception (99.6% training, 98.77% validation) and slightly higher average accuracy of

SVM and softMax classifiers on validation data (99.17% for InceptionV3 vs. 98.98% for Xception) demonstrated its appropriateness for this particular problem. The average accuracy of three feature extractor model on softMax and SVM classifier is shown in Table IV. In addition to determining the better feature extractor model, the better classifier for this particular problem was also determined. As illustrated in Table III SVM classifier outperforms softMax classifier when used with InceptionV3 features. It achieved Higher accuracy 99.9% on training, 99.34% on validation) compared to softMax classifier (which has 98.21% validation accuracy). Overall, based on the two experiments, the InceptionV3 model was the better pre-trained model to serve as the better feature extractor to build our proposed model due to its higher average accuracy of softmax and SVM classifier compared to Xception and VGG16 features.

TABLE III: Feature extractor performance using SVM classifier on the custom datasets

Model	Training accuracy	Validation accuracy
VGG16	96.44	94.6
InceptionV3	99.9	99.34
Xception	99.6	98.77

TABLE IV: Feature extractor average accuracy of softmax and SVM classifier on the validation set

Model	Validation accuracy		
	softMax	SVM	Avg accuracy
VGG16	95.3%	94.6%	94.95
InceptionV3	99%	99.34%	99.17
Xception	99.2%	98.77%	98.98

TABLE V: Hyperparameter tuning results of SVM classifier using Bayesian optimization

Hyperparameter	Sample-space	Tuned value
C (Regularization)	[0.8, 1.0, 2.0, 3.0]	2.0
Kernel Type	[linear, poly, rbf]	linear

B. The Proposed Model Performance

After determining a better feature extractor, we did an experiment to tune the hyper-parameters of SVM classifier such as kernel function and, regularization parameter "C" using Bayesian optimization technique. To tune these hyper-parameters, 2000 medicinal plant leaf images, 5 fold cross validation, batch size of 32 and 10 epochs are used. We divided our datasets into a ratio of 80:20 of training and testing sets respectively. The tuned hyperparameter values are shown in Table V.

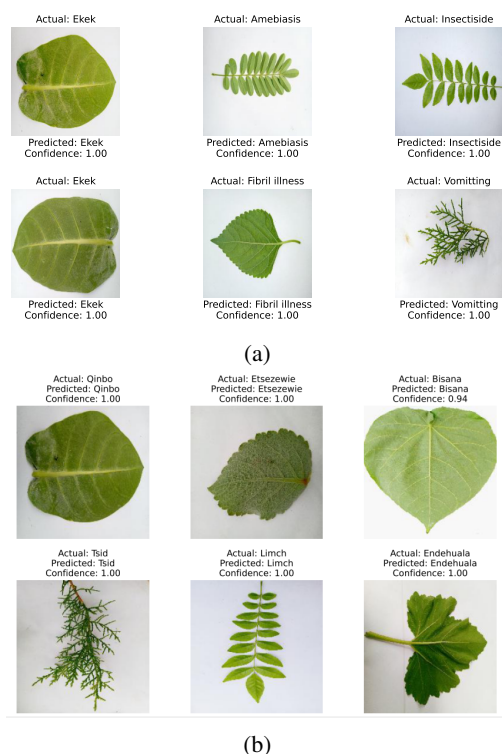


Fig. 4: (a) therapeutic use prediction (b) Medicinal plant species prediction

V. CONCLUSION AND FUTURE WORK

The study focuses on predicting therapeutic uses of indigenous medicinal plants in Ethiopia, preserving and disseminating local knowledge while conserving these resources. Using VGG16, InceptionV3, and Xception deep learning models, the study extracted features from medicinal leaf

datasets. InceptionV3 demonstrated superior performance with validation accuracies of 99.34% (SVM) and 99.17% average accuracy of (softMax and SVM), confirming its selection as the base model. SVM, optimized via Bayesian optimization with InceptionV3 features, achieved a testing accuracy of 99.81% and F1 score of 98.1%, highlighting its efficacy for target prediction. Future research should explore multitask learning for broader applications in medicinal plant research

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